

II-KM – 2-local Cube27 completion, eighth Gauss Fredholm, sharper codimension-three Arb localization, third native Golden H^5 , and full recurrence atlas 45–115

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Abstract

II-KM freezes II-KL under the append-only HT+NT discipline. It localizes the Cube27 augmentation filtration at 2, closes exact I -adic completion on the finite filtered modules while refusing an unsupported splitting of the completed Rees algebra, computes an eighth Gauss transfer trace/Fredholm coefficient, strictly narrows both regular codimension-three return crossings at 256-bit Arb precision, obtains the third native Golden H^5/B^5 closure (type 3, order 32, dimension 56), and repairs the recurrence atlas to a full exact census for every $45 \leq k \leq 115$. The recurrence reflection law is sharpened to a general equal-order collision theorem for even k and odd v . Missing D_5 , historical $p = 5$, Stone and Fibonacci source packets remain explicitly gated.

1 Frozen semantics and checkpoint

Truth states remain {PASS, ZERO, SEPARATED, REFUTED, NT}; mergeability remains an independent axis. Retrieval, numerical replay, theorem application and project-value promotion are separate evidence channels. II-KL enters at (1432, 97, 450, 294, 87) with merge counts (675, 456).

2 Cube27: 2-local filtration and exact completion

The actual relation matrices reproduce

$$\mathrm{gr}_{I,(2)}^1 R \simeq (\mathbb{Z}/2)^2 \oplus \mathbb{Z}/4, \quad \mathrm{gr}_{I,(2)}^2 R \simeq (\mathbb{Z}/2)^5 \oplus \mathbb{Z}/4,$$

$$\boxed{\mathrm{gr}_{I,(2)}^n R \simeq (\mathbb{Z}/2)^6 \oplus \mathbb{Z}/4 \quad (n \geq 3).}$$

The filtration $0 \subset 2M \subset M$ has $(\dim M/2M, \dim 2M) = (3, 1), (6, 1), (7, 1)$ in degrees $1, 2, \geq 3$, and all 60 serialized degree-one operators preserve it.

On the declared Cube27 carrier, the integral ring is finite free of rank 16 over $\mathbb{Z}$; hence its localization $R_{(2)}$ is Noetherian. Completion exactness for finite modules over a Noetherian ring (Stacks Project, Algebra Lemma 10.97.2, tag 00MB) therefore applies to

$$0 \rightarrow I_{(2)}^{n+1} \rightarrow I_{(2)}^n \rightarrow \mathrm{gr}_I^n(R)_{(2)} \rightarrow 0,$$

giving

$$\boxed{0 \rightarrow \widehat{I_{(2)}^{n+1}} \rightarrow \widehat{I_{(2)}^n} \rightarrow \mathrm{gr}_I^n(R)_{(2)} \rightarrow 0.}$$

The conductor class remains nonzero at 2: $\dim_{\mathbb{F}_2} A_2/2A_2 = 13$ and the frozen detector has value 1. It vanishes after inverting 2. Exact completion does *not* provide compatible splittings; “period-two associated graded implies period-two filtered completion” is REFUTED as an inference, while a split/polynomial completed-Rees presentation remains SEPARATED.

3 Gauss and codimension-three return

For the inherited trace-class Gauss carrier, $6^8 = 1,679,616$ exact branch words are grouped into 53,613 matrix-trace classes. At 192-bit Arb precision,

$$\boxed{0.1690543204202985 < \mathrm{tr}(L^8) < 39.00586060099235}$$

and Newton's identity yields

$$-5.038088123389629 < d_8 < 2.113462121846312.$$

The d_8 interval crosses zero. The reduced resolvent exists abstractly from II-KL, but a certified numerical norm and a proof-grade numerical $P'''(0)$ remain **SEPARATED**.

At 256 bits the two regular crossings contract strictly to

$$c_{\sqrt{5}/4} \in [0.8726469017985619, 0.8732370528298088],$$

$$c_{\pi/6} \in [0.9226384987737402, 0.9232502005793504].$$

Their derivative enclosures remain strictly negative. The alternating source-bound C_8 bad-stratum incidence is retained, but no cross-singular transition arrows are recovered; the true compactified germinal leaf groupoid remains **SEPARATED**.

4 Golden-Hodge: third native H^5/B^5 closure

II-KM treats the actual type-3 stabilizer of order 32. It finds 29 elementary-abelian E_8 subgroups. The candidate family contains 3884 products built only from actual native H^1, H^2, H^3 representatives. Each selected E_8 contributes an independent actual block

$$B^5(E) = \text{im}(\delta : C^4(E; \mathbb{F}_2) \rightarrow C^5(E; \mathbb{F}_2)).$$

The joint candidate rank progresses through 21, 27, 33, 34, $\dots$, 49, 53, 56 and reaches the independent upper bound after 17 subgroups and

$$557056 \text{ streamed } C^5(E_8) \text{ rows.}$$

Therefore

$$\dim H_{\text{native}}^5(\text{type } 3) = 56, \quad \text{displayed native products span } H^5 : \text{PASS.}$$

This is the third native degree-five closure, after types 2 and 4. It is not transferred to types 0, 1, 5, 6. The free $\mathbb{F}_2[H]$ -Wall degree-four producer remains **SEPARATED**.

5 Recurrence: a full exact atlas and a reflection collision theorem

For

$$f_{k,v}(x) = x^k - x^v - 1 \in \mathbb{F}_3[x], \quad 1 \leq v < k,$$

II-KM repairs full orders 91, 93, 95 (276 records), scans all orders 106–115 (1095 records), and merges the exact predecessor scopes into

$$\sum_{k=45}^{115} (k-1) = 5609 \text{ factor-certified families.}$$

No state-space 3^k enumeration is used.

For even k and odd v ,

$$f_{k,k-v}(x) = -x^k f_{k,v}(-1/x),$$

so $C(k, k-v)$ is similar to $-C(k, v)^{-1}$. Let $N = \text{ord } C(k, v)$. Since $\det C = -1$, N is even. If $-I \notin \langle C \rangle$, the reflected order is $\text{lcm}(2, N) = N$. If $C^{N/2} = -I$, determinants force $4 \mid N$, hence $\gcd(N, N/2 - 1) = 1$ and again the reflected order is N . Thus

$$\text{ord } C(k, k-v) = \text{ord } C(k, v) \quad (k \text{ even, } v \text{ odd}).$$

The full atlas verifies 691/691 unordered applicable pairs with zero violations.

The exact quotient has 1130 collision classes, including 297 cross-order classes. The largest class has size 11 at matrix order 6461081889226673298932240. The divisibility census gives

$$k \mid \text{ord } C : 621, \quad v \mid \text{ord } C : 1311, \quad (k+v) \mid \text{ord } C : 494.$$

All 35 even fixed centers satisfy

$$\text{ord } C = 4k, \quad h_{\text{anti}} = 2k, \quad \gcd(\text{ord } C, k+v) = k/2, \quad (k+v) \nmid \text{ord } C.$$

The odd-order global-antiperiod sector is **ZERO** on the entire full atlas. Likewise there are no finite-scope hits of $\text{ord } C = k$ or $\text{ord } C = k+v$.

6 Source gates, ledger and safe continuation

Fresh declared Library/GitHub searches recover no new continuous rank-three D_5 transition packet and no full historical $p = 5$ local Tate byte packet. These are scoped retrieval observations only. Stone still lacks the actual project d, θ, K_Z, C and framing/residue provenance; Fibonacci still lacks an instantiated carrier/localization/cofinal-basis/trace-table packet. Hence

ABI-complete actual packets = 0, promoted values = 0, 37 gates active.

The append-only checkpoint closes at

1457 PASS, 100 ZERO, 460 SEPARATED, 295 REFUTED, 90 NT,

687 PROVEN-MERGEABLE, 464 PROVEN-NOT-MERGEABLE.

The deep verifier closes 11 families with 9115 explicit assertions, 8861 witness records and 2463320 coverage units. Observational continuity is PASS_298; exhaustive historical replay remains NT_NOT_EXHAUSTIVE.

Safe successor.

II-KN – Cube27 completed-Rees extension-class splitting/no-splitting and topological-algebra presentation audit; Gauss ninth Fredholm trace/coefficient plus quantitative reduced-resolvent enclosure or proof-grade third-pressure value; codimension-3 cross-singular transition/holonomy packet acquisition and compactified leaf-groupoid cocycle audit; Golden fourth native H5/B5 closure among types 0,1,5,6 or trusted free F2[H]-Wall degree-four producer; post-A1BJ/A1BK D5 continuous transition-source ingestion; historical Pisano $p=5$ local Tate packet acquisition; recurrence full orders 116–125 with collision/divisibility-stratum refinement beyond the equal-order reflection theorem; plus external Stone/Fibonacci ABI-complete packet execution.

References

- [1] II-KL frozen formal bundle and authority registry.
- [2] The Stacks Project, Algebra, Lemma 10.97.2 (tag 00MB), completion for Noetherian rings.
- [3] D. B. A. Epstein and E. Vogt, A Counterexample to the Periodic Orbit Conjecture in Codimension 3, Ann. of Math. 108 (1978), 539–552; project copy hash-locked in the predecessor lineage.